

PRACTICE SHEET

1. If $f(x) = A \sin$ and $\int_0^1 f(x) dx = \frac{2A}{\pi}$, then what is the value of B?
 (a) $2/\pi$ (b) $4/\pi$
 (c) 0 (d) 1
2. If m and n are integers, then what is the value of $\int_0^\pi \sin mx \sin nx dx$, if $m \neq n$?
 (a) 0 (b) $\frac{1}{m+n}$
 (c) $\frac{1}{m-n}$ (d) mn
3. What is the value of $\int_0^1 (x-1)e^{-x} dx$?
 (a) 0 (b) e
 (c) $\frac{1}{e}$ (d) $-\frac{1}{e}$
4. $\int_{\ln 2}^x (e^x - 1) dx = \ln \frac{3}{2}$, then what is the value of x ?
 (a) e^2 (b) $\frac{1}{e}$
 (c) $\ln 4$ (d) 1
5. If $\int_{-3}^2 f(x) dx = \frac{7}{3}$ and $\int_{-3}^9 f(x) dx = -\frac{5}{6}$, then what is value of $\int_{-3}^9 f(x) dx = -\frac{5}{6}$?
 (a) $-\frac{19}{6}$ (b) $\frac{19}{6}$
 (c) $\frac{3}{2}$ (d) $-\frac{3}{2}$
6. You are examine these two statements carefully and select the answer.
Assertion (A) : $\int_0^\pi \sin^7 x dx = 2 \int_0^{\pi/2} \sin^7 x dx$
Reason (R) : $\sin^7 x$ is an odd function
 (a) Both A and R are individually true and R is the correct?
 (b) Both A and R are individually true but R is not the correct explanation A
 (c) A is true but R is false
 (d) A is false but R is true
7. What is the value of integral $\int_0^1 x(1-x)^9 dx$?
 (a) $\frac{1}{110}$ (b) $\frac{1}{111}$
- (c) $\frac{1}{112}$ (d) $\frac{1}{119}$
8. What is the value of $\int_{-1}^1 x |x| dx$?
 (a) 2 (b) 1
 (c) $\frac{1}{4}$ (d) 0
9. What is the value of $\int_0^1 \cos^8 x dx$?
 (a) $\frac{35\pi}{256}$ (b) $\frac{70}{256}$
 (c) $\frac{16}{35}$ (d) $\frac{8\pi}{35}$
10. What is $\int_a^b \frac{\log x}{x} dx$ equal to?
 (a) $(1, 2) \log(ab)$. $\log(b/a)$
 (b) $\log b / \log a$
 (c) $\log(b/a)$
 (d) $(1/2) \log[(a+b)/ab]$
11. What is the value of $\int_0^1 x e^{x^2} dx$?
 (a) $\frac{(e-1)}{2}$ (b) $e^2 - 1$
 (c) $2(e-1)$ (d) $e - 1$
12. The value of $\int_{-2}^2 (ax^3 + bx + c) dx$ depends on which of the following?
 (a) Value of x only
 (b) Values of each of a, b and c
 (c) Value of c only
 (d) Value of b only
13. What are the values of p which satisfy the question $\int_0^p (3x^2 + 4x - 5) dx = p^3 - 2$?
 (a) $1/2$ and 2 (b) $-1/2$ and 2
 (c) $1/2$ and -2 (d) $-1/2$ and -2
14. What is the value of $\int_0^{\pi/2} \log(\tan x) dx$?
 (a) 0 (b) 1
 (c) -1 (d) $\pi/4$
15. What is $\int_0^1 x(1-x)^n dx$ equal to?
 (a) $\frac{1}{n(n+1)}$ (b) $\frac{1}{(n+1)(n+2)}$
 (c) 1 (d) 0
16. What is the value of the integral $\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$?
 (a) $e \left(\frac{e}{2} - 1 \right)$ (b) $e(e-1)$

- (c) $e - \frac{1}{e}$ (d) 0
17. What is the value of the integral $\int_{-1}^1 |x| dx$?
 (a) 1 (b) 0
 (c) 2 (d) -1
18. What is the value of $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin x \cos x}$?
 (a) $2\log \sqrt{3}$ (b) $\log \sqrt{3}$
 (c) $2\log 3$ (d) $4\log 3$
19. What is $\int_{-\pi/4}^{\pi/4} \tan^3 x dx$ equal to?
 (a) $\sqrt{3}$ (b) $\frac{1}{3}$
- (c) $\frac{1}{2}$ (d) 0
20. What is $\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx$?
 (a) π (b) $\frac{\pi}{2}$
 (c) $\frac{\pi}{4}$ (d) 0
21. What is $\int_0^1 x(1-x)^n dx$ equal to?
 (a) $\frac{1}{n(n+1)}$ (b) $\frac{1}{(n+1)(n+2)}$
 (c) 1 (d) 0

ANSWER KEYS

1.	c	2.	a	3.	d	4.	c	5.	a	6.	b	7.	a	8.	d	9.	a	10.	a
11.	a	12.	c	13.	a	14.	a	15.	b	16.	a	17.	a	18.	b	19.	d	20.	c
21.	b																		

Solutions

Sol.1. (c)

Given function $f(x) = A \sin\left(\frac{\pi x}{2}\right) + B$

Differentiating w.r.t. x

$$f'(x) = A \cos\left(\frac{\pi x}{2}\right) \cdot \frac{\pi}{2}$$

$$f'\left(\frac{1}{2}\right) = \sqrt{2} = A \left(\cos\frac{\pi}{4}\right) = \frac{\pi}{2} = A \cdot \frac{1}{\sqrt{2}} \cdot \frac{\pi}{2}$$

$$\Rightarrow A = \frac{(\sqrt{2} \times \sqrt{2}) \times 2}{\pi} = \frac{4}{\pi}$$

$$\text{Now, } \int_0^1 f(x) dx = \frac{2A}{\pi}$$

$$\Rightarrow \int_0^1 \left\{ A \sin\left(\frac{\pi x}{2}\right) + B \right\} dx = \frac{2 \times 4}{\pi^2}$$

$$\Rightarrow \left[-A \cos\frac{\pi x}{2} \cdot \frac{2}{\pi} + Bx \right]_0^1 = \frac{8}{\pi^2}$$

$$\Rightarrow -\frac{4}{\pi} \cdot \frac{2}{\pi} \cos\frac{\pi}{2} + B + \frac{4}{\pi} \cdot \frac{2}{\pi} \cos 0 = \frac{8}{\pi^2}$$

$$\Rightarrow B + \frac{8}{\pi^2} = \frac{8}{\pi^2} \Rightarrow B = 0$$

Sol.2. (a)

The given integral

$$\int_0^{\pi} \sin mx \cdot \sin nx \, dx$$

$$= \int_0^{\pi} \sin m(\pi - x) \cdot \sin n(\pi - x) dx$$

$$= \int_0^{\pi} \sin mx \cdot \sin nx \, dx$$

$$\text{So, } \int_0^{\pi} \sin mx \cdot \sin nx \, dx$$

$$= 2 \int_0^{\pi/2} \sin mx \cdot \sin nx \, dx$$

$$\int_0^{\pi/2} \frac{1}{2} [\cos(mx - nx) - \cos(mx + nx)] \, dx$$

$$= \frac{\pi/2}{0} [\cos(m-n)x - \cos(m+n)x] dx$$

$$= \left[\frac{\sin(m-n)x}{m-n} - \frac{\sin(m+n)x}{m+n} \right]_0^{\pi/2} = 0$$

Sol.3. (d)

Given integral is $I = \int_0^1 (x-1)e^{-x} dx$

Integrating by parts taking $(x-1)$ as first function

$$\text{We get, } I = \left[(x-1) \left(-e^{-x} \right) \right]_0^1 - \int_0^1 1 \cdot \left(-e^{-x} \right) dx$$

$$= - (1-1)1/e + (-1)e^0 + \left[-e^{-x} \right]_0^1 = -1 - \frac{1}{e} + 1 = -\frac{1}{e}$$

Sol.4. (c)

$$\text{Let } I = \int_{\ln 2}^x \frac{1}{e^x - 1} dx$$

$$= \int_{\ln 2}^x \frac{1}{e^x - 1} dx$$

$$\text{Put } e^x - 1 = t \Rightarrow e^x = t + 1$$

$$e^x dx = dt \Rightarrow dx = \frac{dt}{e^x}$$

$$\text{or, } dx = \frac{dt}{t+1}$$

$$\text{when } x = \ln 2, t = e^{\ln 2} - 1 = 2 - 1 = 1$$

$$\text{and } I = \int_1^t \frac{1}{t(t+1)} dt$$

breaking into partial fractions.

$$\frac{1}{t(t+1)} = \frac{t}{t} - \frac{1}{t+1}$$

$$\text{and } I = \int_1^t \left(\frac{1}{t} - \frac{1}{t+1} \right) dt = \left[\log_e t - \log_e (t+1) \right]_1^t$$

$$\text{or } I = \left[\log_e \frac{t}{t+1} \right]_1^t = \log \frac{t}{t+1} - \log_e \frac{1}{2}$$

$$= \text{so, } \frac{2t}{t+1} = \log_e \frac{3}{2}$$

$$\left[\text{since, } \int_{\ln 2}^x \frac{1}{e^x - 1} dx = \log_e \frac{3}{2} \right]$$

$$= \text{so, } \frac{2t}{t+1} = \frac{3}{2}$$

$$\text{or, } 4t = 3t + 3 \Rightarrow t = 3$$

$$\text{so, } e^x - 1 = 3, e^x = 4 \Rightarrow x = \ln 4$$

Sol.5. (a)

$$\text{Value of the integral } \int_2^9 f(x) dx$$

$$\int_{-3}^9 f(x) dx - \int_{-3}^2 f(x) dx$$

$$\text{Given } \int_{-3}^9 f(x) dx = \frac{-5}{6} \text{ and } \int_{-3}^2 f(x) dx = \frac{7}{3}$$

Putting these values in equation (i)

$$\int_{-3}^9 f(x) dx = \frac{-5}{6} - \frac{7}{3} = \frac{19}{6}$$

Sol.6. (b)

$$\int_0^x \sin^7 x dx = 2 \int_0^{x/2} \sin^7 x dx$$

$\sin x$ is an odd function and for an odd function

$$\int_0^a f(x) dx = 2 \int_0^{a/2} f(x) dx$$

$$\text{Hence, } \int_0^x \sin^7 x dx = 2 \int_0^{x/2} \sin^7 x dx \text{ is true}$$

Sol.7. (a)

Let the given integral be,

$$I = \int_0^1 x(1-x)^9 dx$$

Put $1-x = t \Rightarrow dx = -dt$ and $x = 1-t$
When $x = 0, t = 1$ and when $x = 1, t = 0$

$$\Rightarrow I = \int_1^0 (1-t)t^9 (-dt)$$

$$= - \int_0^1 (-t^{10} + t^9) dt = \int_0^1 (-t^{10} + t^9) dt$$

$$\left[-\frac{t^{11}}{11} + \frac{t^{10}}{10} \right]_0^1 = \frac{-1}{11} + \frac{1}{10} = \frac{-10+11}{110} = \frac{1}{110}$$

Sol.8. (d)

Let $f(x) = x|x|$

$$f(-x) = -x|-x| = -x|x| = -f(x)$$

$\Rightarrow x|x|$ is an odd function

$$\text{Hence } \int_{-1}^1 x|x| dx = 0$$

Sol.9. (a)

$$\text{Let, } I = \int_0^{\pi/2} \cos^8 x dx$$

Given integral can be also be written as:

$$I = \int_0^{\pi/2} \sin^0 x \cos^8 x dx, \text{ which is known as}$$

Gamma function.

$$\text{Solution is: } \int_0^{\pi/2} \sin^m x \cos^n x dx$$

$$= \frac{[(m-1)(m-3) \dots 2 \text{ or } 1][(n-1)(n-3) \dots 2 \text{ or } 1]}{[(m+n)(m+n-2) \dots 2 \text{ or } 1]} \cdot \frac{\pi}{2}$$

If m and n both are even then RHS should be multiplied by $\pi/2$ here, $m = 0, N = 8$

$$\Rightarrow I = \frac{(8-1)(8-3)(8-5) \dots (8-7)}{8 \cdot (8-2)(8-4)(8-6)} \cdot \frac{\pi}{2}$$

$$= \frac{7 \cdot 5 \cdot 3 \cdot 1}{8 \cdot 6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{35\pi}{256}$$

Sol.10. (a)

Let the given integer be

$$I = \int_a^b \frac{\log x}{x} dx$$

Put $\log x = t$ and $\frac{dx}{x} = dt$ when $x = a, t = \log a$ and

if $x = b, t = \log b$

$$\therefore I = \int_{\log a}^{\log b} t dt = \left[\frac{t^2}{2} \right]_{\log a}^{\log b}$$

$$= \frac{1}{2} [(\log b)^2 - (\log a)^2]$$

$$= \frac{1}{2} [(\log b + \log a)(\log b - \log a)]$$

$$= \frac{1}{2} \log(ab) \log\left(\frac{b}{a}\right)$$

Sol.11. (a)

$$\text{Let } I = \int_0^1 x e^{x^2} dx$$

Let $x^2 = t$

$$\Rightarrow 2x \, dx = dt$$

$$\Rightarrow x \, dx = \frac{dt}{2}$$

when $x=0$, $t=0$ then $x=1$, $t=1$

$$\Rightarrow I = \frac{1}{2} \int_0^1 e^t \, dt = \frac{1}{2} \left[e^t \right]_0^1$$

$$= \frac{1}{2} \left[e^{x^2} \right]_0^1 = \frac{1}{2} \left[e - e^0 \right] = \frac{e-1}{2}$$

Sol.12. (c)

$$\int_{-2}^2 (ax^3 + bx + c) \, dx$$

$$= \left[\frac{ax^4}{4} + \frac{bx^2}{2} + \frac{cx}{1} \right]_{-2}^2$$

$$= \left[\frac{a(16)}{4} + \frac{b(4)}{2} + 2c \right] - \left[\frac{a(16)}{4} + \frac{b(4)}{2} - 2c \right] = 4c$$

So, the value of given integral depends on the value of c only

Sol.13. (a)

Given equation,

$$\int_0^p (3x^2 + 4x - 5) \, dx = p^3 - 2$$

$$\Rightarrow \left[\frac{3x^3}{3} + \frac{4x^2}{2} - 5x \right]_0^p = p^3 - 2$$

$$\Rightarrow p^3 + 2p^2 - 5p = p^3 - 2$$

$$\Rightarrow 2p^2 - 5p + 2 = 0$$

$$\Rightarrow 2p^2 - 4p - p + 2 = 0$$

$$\Rightarrow 2p(p-2) - 1(p-2) = 0$$

$$\Rightarrow (p-2)(2p-1) = 0$$

$$\Rightarrow p-2=0, 2p-1=0$$

Sol.14. (a)

$$\text{Let } I = \int_0^{\pi/2} \log(\tan x) \, dx \dots (i)$$

$$\text{and } I = \int_0^{\pi/2} \log \left\{ \tan \left(\frac{\pi}{2} - x \right) \right\} \, dx$$

[By the property of definite integral which says

$$\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx \dots (ii)$$

$$\Rightarrow I = \int_0^{\pi/2} \log(\cot x) \, dx$$

By adding equation (i) and (ii), we get

$$2I = \int_0^{\pi/2} \log(\tan x) \, dx + \int_0^{\pi/2} \log(\cot x) \, dx$$

$$2I = \int_0^{\pi/2} \log(\tan x) \, dx + \int_0^{\pi/2} \log(\cot x) \, dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \log(\tan x \cot x) \, dx$$

$$[\log m + \log n = \log(mn)]$$

$$= \int_0^{\pi/2} \log \left(\tan x \cdot \frac{1}{\tan x} \right) \, dx = \int_0^{\pi/2} \log 1 \, dx = 0$$

$$\Rightarrow I = 0$$

Sol.15. (b)

$$\text{Let } I = \int_0^1 x(1-x)^n \, dx$$

$$\text{Put } 1-x = t \Rightarrow dx = -dt$$

$$\text{When } x=0 \text{ then } t=1$$

$$\text{When } x=1 \text{ then } t=0$$

$$\therefore I = - \int_1^0 (1-t)t^n \, dt = \int_0^1 (t^n - t^{n+1}) \, dt$$

$$= \left[\frac{t^{n+1}}{n+1} - \frac{t^{n+2}}{n+2} \right]_0^1$$

$$= \frac{1}{n+1} - \frac{1}{n+2} = \frac{1}{(n+1)(n+2)}$$

Sol.16. (a)

$$\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) \, dx = \left(\frac{e^x}{x} \right)_1^2 = e \left(\frac{e}{2} - 1 \right)$$

Sol.17. (a)

$$\int_{-1}^1 |x| \, dx = 2 \int_0^1 x \, dx = 1$$

Sol.18. (b)

$$\int_{\pi/6}^{\pi/4} \frac{dx}{\sin x \cos x} = 2 \int_{\pi/6}^{\pi/4} \frac{dx}{2 \sin x \cos x} = 2 \int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x}$$

$$= 2 \int_{\pi/6}^{\pi/4} \operatorname{cosec} 2x \, dx = 2 [\log \tan x]_{\pi/6}^{\pi/4} \cdot \frac{1}{2} = \log \sqrt{3}$$

Sol.19. (d)

$$\int_{-\pi/4}^{\pi/4} \tan^3 x \, dx = 0 \quad (\text{odd function})$$

Sol.20. (c)

$$I = \int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} \, dx$$

$$\left[\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx \right]$$

$$I = \int_0^{\pi/2} \frac{\cos^3 x}{\sin^3 x + \cos^3 x} \, dx$$

$$2I = \int_0^{\pi/2} \frac{\sin^3 x + \cos^3 x}{\sin^3 x + \cos^3 x} \, dx = \int_0^{\pi/2} 1 \, dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Sol.21. (b)

$$\int_0^1 x(1-x)^n \, dx$$

$$dx \left[\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx \right]$$

$$= \int_0^1 (1-x)(x)^n \, dx = \int_0^1 (x^n - x^{n+1}) \, dx$$

$$= \int_0^1 (x^n - x^{n+1}) \, dx = \left[\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2} \right]_0^1$$

$$= \left[\frac{1}{n+1} - \frac{1}{n+2} \right] = \frac{1}{(n+1)(n+2)}$$

NDA PYQ

1. If $f(x)$ is an even function, then what is $\int_0^{\pi} f(\cos x) dx$ equal to?
 (a) 0 (b) $\int_0^{\pi/2} f(\cos x) dx$
 (c) $2 \int_0^{\pi/2} f(\cos x) dx$ (d) 1
[NDA (I) - 2011]
2. What is $\int_0^{\pi} \frac{dx}{1+2\sin^2 x}$ equal to?
 (a) π (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{\sqrt{3}}$ (d) $\frac{2\pi}{\sqrt{3}}$
[NDA (I) - 2011]
3. If $I_n = \int_0^{\pi/4} \tan^n x dx$, then what is $I_n + I_{n-2}$ equal to?
 (a) $\frac{1}{n}$ (b) $\frac{1}{(n-1)}$
 (c) $\frac{n}{(n-1)}$ (d) $\frac{1}{(n-2)}$
[NDA (I) - 2011]
4. If $\int_1^2 \{k^2 + (4-4k)x + 4x^3\} dx \leq 12$, then which one of the following is correct?
 (a) $k=3$ (b) $0 \leq k < 3$
 (c) $k \leq 4$ (d) $k=0$
[NDA (II)-2011]
5. What is the value of $\int_{-\pi/2}^{\pi/2} |\sin x| dx$?
 (a) 2 (b) 1
 (c) π (d) 0
[NDA (I) - 2012]
6. What is the value of $\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$?
 (a) $\frac{\pi^2}{8}$ (b) $\frac{\pi^2}{32}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{8}$
[NDA (I) - 2012]
7. What is the value of $\int_0^{\pi/2} \sin 2x \ln(\cot x) dx$?
 (a) 0 (b) $\pi \ln 2$
 (c) $-\pi \ln 2$ (d) $\frac{\pi \ln 2}{2}$
[NDA (II) - 2012]
8. What is the value of $\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$?
 (a) $\frac{\pi^2}{8}$ (b) $\frac{\pi^2}{32}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{8}$
[NDA (II) - 2012]
9. What is $\int_{-1}^1 x |x| dx$ equal to?
 (a) 2 (b) 1
 (c) 0 (d) -1
[NDA-2012(2)]
10. What is $\int_0^1 x e^x dx$ equal to?
 (a) 1 (b) -1
 (c) 0 (d) e
[NDA (I) - 2013]
11. What is $\int_0^2 \frac{dx}{x^2+4}$ equal to?
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{8}$ (d) $\frac{\pi}{3}$
[NDA (I) - 2013]
12. What is $\int_{-a}^a (x^3 + \sin x) dx$ equal to?
 (a) a (b) 2a
 (c) 0 (d) 1
[NDA (I) - 2013]
13. What is $\int_{-\pi/6}^{\pi/6} \frac{\sin^5 x \cos^3 x}{x^4} dx$ equal to?
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{8}$ (d) 0
[NDA (I) - 2013]
14. What is $\int_0^2 e^{\log x} dx$ equal to?
 (a) 1 (b) 2
 (c) 4 (d) None of these
[NDA (II) - 2013]
15. What is $\int_1^2 \log x dx$ equal to?
 (a) $\log e^2$ (b) 1
 (c) $\log_e \left(\frac{4}{e}\right)$ (d) $\log_e \left(\frac{e}{4}\right)$
[NDA (II) - 2013]

16. What is $\int_{-\pi/2}^{\pi/2} x \sin x \, dx$ equal to?
 (a) 0 (b) 2
 (c) -2 (d) π

[NDA (I) - 2014]

17. What is $\int_0^{\pi/2} \ln(\tan x) \, dx$ equal to?
 (a) $\ln 2$ (b) $-\ln 2$
 (c) 0 (d) None of these

[NDA (I) - 2014]

18. What is $\int_0^1 \frac{e^{\tan^{-1}x} \, dx}{1+x^2}$ equal to?
 (a) $e^{\frac{\pi}{4}} - 1$ (b) $e^{\frac{\pi}{4}} + 1$
 (c) $e - 1$ (d) e

[NDA (I) - 2014]

Directions (for next two):

$$I_1 = \int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}} \quad \text{and} \quad I_2 = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x} \, dx}{\sqrt{\sin x} + \sqrt{\cos x}}$$

19. What is $I_1 - I_2$ equal to?
 (a) 0 (b) $2I_1$
 (c) π (d) None of these

[NDA (I) - 2014]

20. What is I_1 equal to?
 (a) $\frac{\pi}{24}$ (b) $\frac{\pi}{18}$
 (c) $\frac{\pi}{12}$ (d) $\frac{\pi}{6}$

[NDA (I) - 2014]

Directions (for next two):

Consider the integral $I = \int_0^{\pi} \ln(\sin x) \, dx$

21. What is $\int_0^{\pi/2} \ln(\sin x) \, dx =$
 (a) $4I$ (b) $2I$
 (c) I (d) $\frac{I}{2}$

[NDA (II) - 2014]

22. What is $\int_0^{\pi/2} \ln(\cos x) \, dx$ equal to?
 (a) $\frac{I}{2}$ (b) I
 (c) $2I$ (d) $4I$

[NDA (II) - 2014]

23. What is $\int_0^{\pi/2} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$ equal to?
 (a) $2ab$ (b) $2\pi ab$
 (c) $\frac{\pi}{2ab}$ (d) $\frac{\pi}{ab}$

[NDA (II) - 2014]

Directions (for next three): Read the following information carefully and answer the questions given below.
 For the next three solutions consider:

$$I = \int_0^{\pi} \frac{x \, dx}{1 + \sin x}$$

24. What is I equal to?
 (a) $-\pi$ (b) 0
 (c) π (d) 2π

[NDA (II) - 2014]

25. What is $\int_0^{\pi} \frac{(\pi - x) \, dx}{1 + \sin x}$ equal to?
 (a) π (b) $\frac{\pi}{2}$
 (c) 0 (d) 2π

[NDA (II) - 2014]

26. What is $\int_0^{\pi} \frac{dx}{1 + \sin x}$ equal to?
 (a) 1 (b) 2
 (c) 4 (d) -2

[NDA (II) - 2014]

Directions (for next four): Consider the integral $I_m =$

$$\int_0^{\pi} \frac{\sin 2mx}{\sin x} \, dx, \text{ where } m \text{ is a positive integer.}$$

27. What is I_1 equal to?
 (a) 0 (b) $\frac{1}{2}$
 (c) 1 (d) 2

[NDA (I) - 2015]

28. What $I_2 + I_3$ equal to?
 (a) 4 (b) 2
 (c) 1 (d) 0

[NDA (I) - 2015]

29. What is I_m equal to:
 (a) 0 (b) 1
 (c) m (d) $2m$

[NDA (I) - 2015]

30. Consider the following:
 I. $I_m + I_{m-1}$ is equal to 0
 II. $I_{2m} > I_m$
 Which of the above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

[NDA (I) - 2015]

31. The value of: $\int_a^b \frac{x^7 + \sin x}{\cos x} \, dx$ where $a + b = 0$ is:
 (a) $2b - a \sin(b-a)$ (b) $a + 3b \cos(b-a)$
 (c) $\sin a - (b-a) \cos b$ (d) 0

[NDA (II) - 2015]

32. $\int_{-1}^1 x |x| \, dx$ is equal to:
 (a) 0 (b) $\frac{2}{3}$
 (c) 2 (d) -2

[NDA (II) - 2015]

33. If $0 < a < b$, then $\int_a^b \frac{|x|}{x} \, dx$ is equal to?
 (a) $|b| - |a|$ (b) $|a| - |b|$

- (c) $\frac{|b|}{|a|}$ (d) 0

[NDA (II) - 2015]

Direction (for next two) :

Consider the integrals

$$A = \int_0^{\pi} \frac{\sin x dx}{\sin x + \cos x} \text{ and } B = \int_0^{\pi} \frac{\sin x dx}{\sin x - \cos x}$$

34. Which of the following is correct?
(a) $A = 2B$ (b) $B = 2A$
(c) $A = B$ (d) $A = 3B$

[NDA (II) - 2015]

35. What is the value of B ?

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) $\frac{3\pi}{4}$ (d) π

[NDA (II) - 2015]

36. $\int_0^{2\pi} \sin^5\left(\frac{x}{4}\right) dx$ is equal to:

- (a) $\frac{8}{15}$ (b) $\frac{16}{15}$
(c) $\frac{32}{15}$ (d) 0

[NDA (II) - 2015]

37. What is $\int_0^{4\pi} |\cos x| dx$ equal to?

- (a) 0 (b) 2
(c) 4 (d) 8

[NDA (I) - 2016]

For the next two (2) items that follow:

Consider the functions

$$f(x) = xg(x) \text{ and } g(x) = \left[\frac{1}{x} \right]$$

where $[.]$ is the greatest integer function.

38. What is $\int_{\frac{1}{3}}^{\frac{1}{2}} g(x) dx$ equal to?

- (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{5}{18}$ (d) $\frac{5}{36}$

[NDA (I) - 2016]

39. What is $\int_{\frac{1}{3}}^1 f(x) dx$ equal to?

- (a) $\frac{37}{72}$ (b) $\frac{2}{3}$
(c) $\frac{17}{72}$ (d) $\frac{37}{144}$

[NDA (I) - 2016]

40. What is $\int_{-2}^2 x dx - \int_{-2}^2 [x] dx$ equal to, where $[.]$ is the greatest integer function?

- (a) 0 (b) 1
(c) 2 (d) 4

[NDA (I) - 2016]

41. If $\int_{-2}^5 f(x) dx = 4$ and $\int_0^5 \{1+f(x)\} dx = 7$, then what is

$$\int_{-2}^0 f(x) dx \text{ equal to?}$$

- (a) -3 (b) 2
(c) 3 (d) 5

[NDA (I) - 2016]

For the next two (2) items that follow:

$$\text{Given that } a_n = \int_0^{\pi} \frac{\sin^2\{(n+1)x\}}{\sin 2x} dx$$

42. Consider the following statements:

1. The sequence $\{a_{2n}\}$ is in AP with common difference zero.

2. The sequence $\{a_{2n+1}\}$ is in AP with common difference zero

Which of the above statements is/are correct?

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2016(1)]

43. What is $a_{n-1} - a_{n-4}$ equal to?

- (a) -1 (b) 0
(c) 1 (d) 2

[NDA-2016(1)]

44. Let $f(x)$ be a function such that:

$$f\left(\frac{1}{x}\right) + x^3 f'(x) = 0. \text{ What is } \int_{-1}^1 f(x) dx \text{ equal to?}$$

- (a) $2f(1)$ (b) 0
(c) $2f(-1)$ (d) $4f(1)$

[NDA (II) - 2016]

45. If $\int_0^{\pi/2} \frac{dx}{3\cos x + 5} = k \cot^{-1} 2$, then what is the value of k?

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) 1 (d) 2

[NDA (II) - 2016]

46. What is $\int_1^3 |1-x^4| dx$ equal to?

- (a) $-\frac{232}{5}$ (b) $-\frac{116}{5}$
(c) $\frac{116}{5}$ (d) $\frac{232}{5}$

[NDA (II) - 2016]

47. What is $\int_0^{\pi/2} \frac{d\theta}{1+\cos \theta}$ equal to?

- (a) $\frac{1}{2}$ (b) 1
(c) $\sqrt{3}$ (d) None of these

[NDA (I) - 2017]

48. If $f(x)$ and $g(x)$ are continuous functions satisfying $f(x) = f(a-x)$ and $g(x) + g(a-x) = 2$, then what is $\int_0^a f(x) g(x) dx$ equal to?

(a) $\int_0^a g(x) dx$ (b) $\int_0^a f(x) dx$
 (c) $2 \int_0^a f(x) dx$ (d) 0

[NDA (I) - 2017]

49. What is $\int_{e^{-1}}^{e^2} \left| \frac{\ln x}{x} \right| dx$ equal to?

(a) $\frac{3}{2}$ (b) $\frac{5}{2}$
 (c) 3 (d) 4

[NDA (I) - 2017]

50. What is $\int_0^{2\pi} \sqrt{1 + \sin \frac{x}{2}} dx$ equal to?

(a) 8 (b) 4
 (c) 2 (d) 0

[NDA (II) - 2017]

51. The value of: $\int_0^{\pi/4} \sqrt{\tan x} dx + \int_0^{\pi/4} \sqrt{\cot x} dx$ is equal to:

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
 (c) $\frac{\pi}{2\sqrt{2}}$ (d) $\frac{\pi}{\sqrt{2}}$

[NDA (II) - 2017]

52. What is $\int_0^1 x(1-x)^9 dx$ equal to?

(a) 1/110 (b) 1/132
 (c) 1/148 (d) 1/240

[NDA (I) - 2018]

53. If $\int_a^b x^3 dx = 0$ and $\int_a^b x^2 dx = \frac{2}{3}$, then what are the values of a and b respectively?

(a) -1, 1 (b) 1, 1
 (c) 0, 0 (d) 2, -2

[NDA (I) - 2018]

54. What is the value of $\int_{-\pi/4}^{\pi/4} (\sin x - \tan x) dx$?

(a) $-\frac{1}{\sqrt{2}} + \ln\left(\frac{1}{\sqrt{2}}\right)$ (b) $\frac{1}{\sqrt{2}}$
 (c) 0 (d) $\sqrt{2}$

[NDA (I) - 2018]

55. What is $\int_0^{\sqrt{2}} [x^2] dx$ equal to (where [.] is the greatest integer function)?

(a) $\sqrt{2} - 1$ (b) $1 - \sqrt{2}$
 (c) $2(\sqrt{2} - 1)$ (d) $\sqrt{3} - 1$

[NDA (I) - 2018]

56. What is $\int_1^e x \ln x dx$ equal to:

(a) $\frac{e+1}{4}$ (b) $\frac{e^2+1}{4}$

(c) $\frac{e-1}{4}$ (d) $\frac{e^2-1}{4}$

[NDA (I) - 2018]

57. What is $\int_0^{\pi} e^x \sin x dx$ equal to?

(a) $\frac{e^{\pi}+1}{2}$ (b) $\frac{e^{\pi}-1}{2}$
 (c) $e^{\pi}+1$ (d) $\frac{e^{\pi}+1}{4}$

[NDA (I) - 2018]

58. What is $\int_{-1}^1 \left\{ \frac{d}{dx} \left(\tan^{-1} \frac{1}{x} \right) \right\} dx$ equal to?

(a) 0 (b) $-\frac{\pi}{4}$
 (c) $-\frac{\pi}{2}$ (d) $\frac{\pi}{2}$

[NDA (II) - 2018]

59. What is $\int_2^8 |x-5| dx$ equal to?

(a) 2 (b) 3
 (c) 4 (d) 9

[NDA (II) - 2018]

60. What is $\int_a^b [x] dx + \int_a^b [-x] dx$ equal to, where [.] is the greatest integer function?

(a) b-a (b) a-b
 (c) 0 (d) 2(b-a)

[NDA (II) - 2018]

61. What is $\int_0^{\pi/2} |\sin x - \cos x| dx$ equal to?

(a) 0 (b) $2(\sqrt{2}-1)$
 (c) $2\sqrt{2}$ (d) $2(\sqrt{2}+1)$

[NDA (I) - 2019]

62. What is $\int_0^{\pi/2} e^{\sin x} \cos x dx$ equal to?

(a) e + 1 (b) e - 1
 (c) e + 2 (d) e

[NDA (I) - 2019]

Direction (for the next two):

Consider the integrals

$I_1 = \int_0^{\pi} \frac{x dx}{1 + \sin x}$ and $I_2 = \int_0^{\pi} \frac{(\pi - x) dx}{1 - \sin(\pi + x)}$

63. What is the value of I_1 ?

(a) 2π (b) π
 (c) $\pi/2$ (d) 2π

[NDA (II) - 2019]

64. What is the value of $I_1 + I_2$?

(a) 2π (b) π
 (c) $\pi/2$ (d) 2π

[NDA (II) - 2019]

65. What is the value of $\int_0^{\pi/4} (\tan^3 x + \tan x) dx$?

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) 1 (d) 2 [NDA 2020]
66. What is $\int_0^a \frac{f(a-x)}{f(x)+f(a-x)} dx$ equal to?
(a) a (b) 2a
(c) 0 (d) $\frac{a}{2}$ [NDA (I) - 2021]
67. If $\int_0^a [f(x) + f(-x)] dx = \int_{-a}^a g(x) dx$
then what is $g(x)$ equal to?
(a) $f(x)$ (b) $f(-x) + f(x)$
(c) $-f(x)$ (d) none of these [NDA (I) - 2021]
68. If $f(x) = 2^x$, then what is $\int_2^{10} \frac{f'(x)}{f(x)} dx$ equal to?
(a) $4 \ln 2$ (b) $\ln 4$
(c) $\ln 5$ (d) $8 \ln 2$ [NDA (II) - 2021]
69. If $\int_{-2}^0 f(x) dx = k$, then $\int_{-2}^0 |f(x)| dx$ is:
(a) less than k
(b) greater than k
(c) less than or equal to k
(d) greater than or equal to k [NDA (II) - 2021]
70. If $f(x)$ satisfied $f(1) = f(4)$, then what is $\int_1^4 f'(x) dx$ equal to?
(a) -1 (b) 0
(c) 1 (d) 2 [NDA (II) - 2021]
71. What is $\int_0^{\frac{\pi}{2}} e^{\ln(\cos x)} dx$ equal to?
(a) -1 (b) 0
(c) 1 (d) 2 [NDA (II) - 2021]
72. What is $\int_0^{\frac{\pi}{2}} \ln\left(\tan \frac{x}{2}\right) dx$ equal to?
(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2 [NDA (II) - 2021]
73. What is $\int_0^{\frac{\pi}{4}} \frac{dx}{(\sin x + \cos x)^2}$ equal to?
(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
(c) 1 (d) $\frac{3}{2}$ [NDA (I) - 2022]
74. What is $\int_{-2}^{-1} \frac{x}{|x|} dx$ equal to?
(a) -2 (b) -1

- (c) 1 (d) 2 [NDA (I) - 2022]
75. What is $\int_0^1 \ln\left(\frac{1}{x} - 1\right) dx$ equal to?
(a) -1 (b) 0
(c) 1 (d) $\ln 2$ [NDA 2022 (II)]
76. If $\int_0^{\pi/2} (\sin^4 x + \cos^4 x) dx = k$, then what is the value of $\int_0^{20\pi} (\sin^4 x + \cos^4 x) dx$?
(a) k (b) $10k$
(c) $20k$ (d) $40k$ [NDA 2022 (II)]
77. What is $\int_{-\pi/2}^{\pi/2} (e^{\cos x} \sin x + e^{\sin x} \cos x) dx$ equal to?
(a) $\frac{e^2 - 1}{e}$ (b) $\frac{e^2 + 1}{e}$
(c) $\frac{1 - e^2}{e}$ (d) 0 [NDA 2022 (II)]
- Consider the following for the next three (03) items that follow:
Let $f(x) = Pe^x + Qe^{2x} + Re^{3x}$, where, P, Q, R are real numbers. Further $f(0) = 6$, $f(\ln 3) = 282$ and $\int_0^{\ln 2} f(x) dx = 11$.
78. What is the value of Q ?
(a) 1 (b) 2
(c) 3 (d) 4 [NDA - 2023 (1)]
79. What is the value of R ?
(a) 1 (b) 2
(c) 3 (d) 4 [NDA - 2023 (1)]
80. What is $f'(0)$ equal to?
(a) 18 (b) 16
(c) 15 (d) 14 [NDA - 2023 (1)]
- Consider the following for the next three (03) items that follows:
Let $I_1 = \int_0^{\frac{\pi}{2}} \frac{x}{1 + \cos^2 x} dx$ and $I_2 = \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin^2 x} dx$
81. What is the value of $\frac{I_1 + I_2}{I_1 - I_2}$?
(a) 1 (b) π
(c) π^2 (d) $\frac{\pi+2}{\pi-2}$ [NDA - 2023 (1)]
82. What is the value of $8I_1^2$?
(a) π (b) π^2
(c) π^3 (d) π^4 [NDA - 2023 (1)]
83. What is the value of I_2 ?

- (a) $\frac{\pi}{\sqrt{2}}$ (b) $\frac{\pi}{2\sqrt{2}}$
 (c) $\frac{3\pi}{2\sqrt{2}}$ (d) $\frac{\pi}{4\sqrt{2}}$

[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:

Let $I = \int_a^b \frac{|x|}{x} dx$, $a < b$

84. What is I equal to when $a < 0 < b$?
 (a) $a + b$ (b) $a - b$
 (c) $b - a$ (d) $\frac{(a + b)}{2}$

[NDA – 2023 (1)]

85. What is I equal to when $a < b < 0$?
 (a) $a + b$ (b) $a - b$
 (c) $b - a$ (d) $\frac{(a + b)}{2}$

[NDA – 2023 (1)]

Consider the following for the next (02) items that follow

Let $I = \int_{-2\pi}^{2\pi} \frac{\sin^4 x + \cos^4 x}{1 + 3^x} dx$

86. What is $\int_0^{\pi} (\sin^4 x + \cos^4 x) dx$ equal to?
 (a) $\frac{3\pi}{8}$ (b) $\frac{3\pi}{4}$
 (c) $\frac{3\pi}{2}$ (d) 3π

[NDA-2023 (2)]

87. What is I equal to?
 (a) 0 (b) $\frac{3\pi}{4}$
 (c) $\frac{3\pi}{2}$ (d) 3π

[NDA-2023 (2)]

88. What is $\int_0^{8x} |\sin x| dx$ equal to?
 (a) 2 (b) 4
 (c) 8 (d) 16

[NDA-2023 (2)]

Consider the following for the next two (02) items that follow:

Let $f(x) = |x - 1|$, $g(x) = [x]$ and $h(x) = f(x)g(x)$ where $[.]$ is greatest integer function.

89. What is $\int_{-1}^0 h(x) dx$ equal to?
 (a) $-\frac{3}{2}$ (b) -1
 (c) 0 (d) $\frac{1}{2}$

[NDA-2024 (1)]

90. What is $\int_0^2 h(x) dx$ equal to?

- (a) $-\frac{3}{2}$ (b) -1
 (c) 0 (d) $\frac{1}{2}$

[NDA-2024 (1)]

Consider the following for the next item that follow:

Let $3f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x} + 1$

91. What is $8 \int_1^2 f(x) dx$ equal to?
 (a) $\ln(8\sqrt{e})$ (b) $\ln(4\sqrt{e})$
 (c) $\ln 2$ (d) $\ln 2 - 1$

[NDA-2024 (1)]

92. What is $\int_{-1}^1 (3 \sin x - \sin 3x) \cos^2 x dx$ equal to?
 (a) $-\frac{1}{4}$ (b) 0
 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

[NDA-2024 (1)]

93. Let $p = \int_a^b f(x) dx$ and $q = \int_a^b |f(x)| dx$.
 If $f(x) = e^{-x}$, then which one of the following is correct?
 (a) $p = 2q$ (b) $p = -q$
 (c) $4p = q$ (d) $p = q$

[NDA-2024 (1)]

94. What is $\int_0^{\frac{\pi}{2}} \frac{a + \sin x}{2a + \sin x + \cos x} dx$ equal to?
 (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
 (c) 1 (d) 0

[NDA-2024 (1)]

Consider the following for the next two (02) items that follow:

Let $\phi(a) = \int_a^{a+100\pi} |\sin x| dx$

95. What is $\phi(a)$ equal to?
 (a) 0 (b) a
 (c) $100a$ (d) 200

[NDA-2024 (1)]

96. What is $\phi'(a)$ equal to?
 (a) 0 (b) π
 (c) 100 (d) 200

[NDA-2024 (1)]

Direction: Consider the following for the two items given below

Let $f(x) = |x^2 - x - 2|$

97. What is $\int_0^2 f(x) dx$ equal to?
 (a) 0 (b) 1
 (c) $5/3$ (d) $10/3$

[NDA-2024 (2)]

98. What is $\int_1^3 f(x) dx$ equal to?
 (a) 2 (b) 3
 (c) 4 (d) 5

[NDA-2024 (2)]

Direction: Consider the following for the **one** items given below

Let $f(t) = \ln(t + \sqrt{1+t^2})$ and $g(t) = \tan(f(t))$

99. What is $\int_{-\pi}^{\pi} g(t) dt$ equal to ?

- (a) -1 (b) 0
(c) 1/2 (d) 1

[NDA-2024 (2)]

Direction: Consider the following for the **two** items given below

Let $I = \int_0^{\pi/2} \frac{f(x)}{g(x)} dx$, where $f(x) = \sin x$ and

$g(x) = \sin x + \cos x + 1$

100. What is $\int_0^{\pi/2} \frac{dx}{g(x)}$ equal to?

- (a) $\frac{\ln 2}{2}$ (b) $\frac{\ln 2}{4}$
(c) $\ln 2$ (d) $2 \ln 2$

[NDA-2024 (2)]

101. What is I equal to ?

- (a) $\frac{\pi}{4} + \ln 2$ (b) $\frac{\pi}{4} - \ln 2$
(c) $\frac{\pi}{4} - \frac{\ln 2}{2}$ (d) $\frac{\pi}{4} + \frac{\ln 2}{2}$

[NDA-2024 (2)]

Consider the following for the two (02) items that follow:

Let $f(x) = [x^2]$, where $[.]$ is the greatest integer function.

102. What is $\int_{\sqrt{2}}^{\sqrt{3}} f(x) dx$ equal to?

- (a) $\sqrt{3} - \sqrt{2}$ (b) $2(\sqrt{3} - \sqrt{2})$
(c) $3 - \sqrt{2}$ (d) 1

[NDA-2025 (1)]

103. What is $\int_{\sqrt{2}}^2 f(x) dx$ equal to?

- (a) $6 - \sqrt{3} - 2\sqrt{2}$ (b) $6 - \sqrt{3} - \sqrt{2}$
(c) $6 - \sqrt{3} + 2\sqrt{2}$ (d) $6 + \sqrt{3} - 2\sqrt{2}$

[NDA-2025 (1)]

104. What is $\int_n^{n+1} (x - [x]) dx$, where $[.]$ is the greatest integer function and n is natural number?

- (a) $\frac{4n+1}{2}$ (b) $\frac{2n+1}{2}$
(c) $\frac{1}{2}$ (d) 1

[NDA-2025 (2)]

105. If $I_1 = \int_e^{e^2} \frac{dx}{\ln x}$ and $I_2 = \int_1^2 \frac{e^x}{x} dx$, then which one of the following is correct?

- (a) $I_1 - I_2 = 0$ (b) $I_1 + I_2 = 0$
(c) $I_1 - 2I_2 = 0$ (d) $2I_1 - I_2 = 0$

[NDA-2025 (2)]

ANSWER KEY

1.	c	2.	c	3.	b	4.	a	5.	a	6.	b	7.	a	8.	b	9.	c	10.	a
11.	c	12.	c	13.	d	14.	b	15.	c	16.	b	17.	c	18.	a	19.	a	20.	c
21.	d	22.	a	23.	c	24.	c	25.	a	26.	b	27.	a	28.	d	29.	a	30.	a
31.	d	32.	a	33.	a	34.	c	35.	b	36.	c	37.	d	38.	b	39.	a	40.	c
41.	b	42.	c	43.	b	44.	c	45.	b	46.	d	47.	b	48.	b	49.	b	50.	a
51.	d	52.	a	53.	a	54.	c	55.	a	56.	b	57.	b	58.	d	59.	d	60.	b
61.	b	62.	b	63.	b	64.	d	65.	b	66.	d	67.	a	68.	d	69.	d	70.	d
71.	c	72.	a	73.	b	74.	b	75.	b	76.	d	77.	a	78.	b	79.	c	80.	d
81.	d	82.	d	83.	a	84.	a	85.	b	86.	b	87.	c	88.	d	89.	a	90.	d
91.	a	92.	b	93.	d	94.	a	95.	d	96.	a	97.	d	98.	b	99.	b	100.	c
101.	c	102.	b	103.	a	104.	c	105.	a										

Solutions

Sol.1. (c)

$$\int_0^{\pi} f(\cos x) dx = \int_0^{\pi} f(\cos(\pi - x)) dx = \int_0^{\pi} f(-\cos x) dx$$

$$= \int_0^{\pi} f(-\cos x) dx = \int_0^{\pi} f(\cos x) dx$$

$$\text{so } \int_0^{\pi} f(\cos x) dx = 2 \int_0^{\pi/2} f(\cos x) dx$$

Sol.2. (c)

$$\int_0^{\pi} \frac{dx}{1+2\sin^2 x} = 2 \int_0^{\pi/2} \frac{dx}{1+2\sin^2 x}$$

$$= \int_0^{\pi/2} \frac{\sec^2 x dx}{\sec^2 x + 2 \tan^2 x} = \int_0^{\pi/2} \frac{\sec^2 x dx}{1+3 \tan^2 x}$$

put $\tan x = t$

$$= \int_0^{\infty} \frac{dt}{1+3t^2} = \frac{1}{3} \int_0^{\infty} \frac{dt}{\left(\frac{1}{\sqrt{3}}\right)^2 + t^2} = \frac{\pi}{\sqrt{3}}$$

Sol.3. (b)

$$I_n = \int_0^{\pi/4} \tan^n x dx$$

$$I_n + I_{n-2} = \int_0^{\pi/4} \tan^n x dx + \int_0^{\pi/4} \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x dx + \int_0^{\pi/4} \tan^2 x \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x (1 + \tan^2 x) dx = \int_0^{\pi/4} \tan^{n-2} x (\sec^2 x) dx$$

put $\tan x = t$

$$= \int_0^1 t^{n-2} dt = \left[\frac{t^{n-2+1}}{n-2+1} \right]_0^1 = \frac{1}{n-1}$$

Sol.4. (a)

$$\int_1^2 \{k^2 + (4-4k)x + 4x^3\} dx \leq 12,$$

$$\left(k^2 x + (4-4k) \frac{x^2}{2} + 4 \frac{x^4}{4} \right)_1^2 \leq 12$$

$$\left(k^2 x + (2-2k)x^2 + x^4 \right)_1^2 \leq 12$$

$$(k^2 + 3(2-2k) + 15) \leq 12$$

$$(k^2 - 6k + 21) \leq 12$$

$$k^2 - 6k + 9 \leq 0$$

$$(k-3)^2 \leq 0$$

$$k = 3$$

Sol.5. (a)

$$\int_{-\pi/2}^{\pi/2} |\sin x| dx = 2 \quad (\text{by graph})$$

Sol.6. (a)

$$\int_0^1 x e^x dx = [x e^x - e^x]_0^1 = 1$$

Sol.7. (c)

$$\int_0^2 \frac{dx}{x^2+4} = \frac{1}{2} \left[\tan^{-1} \frac{x}{2} \right]_0^2 = \frac{\pi}{8}$$

Sol.8. (c)

$$\int_{-a}^a (x^3 + \sin x) dx = 0 \quad (\text{odd function})$$

Sol.9. (d)

$$\int_{-\pi/6}^{\pi/6} \frac{\sin^5 x \cos^3 x}{x^4} dx = 0 \quad (\text{odd function})$$

Sol.10. (a)

$$I = \int_0^{\pi/2} \sin 2x \cdot \ln(\cot x) dx$$

$$\left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$I = \int_0^{\pi/2} \sin 2x \cdot \ln(\tan x) dx$$

$$2I = \int_0^{\pi/2} \sin 2x \cdot \ln(\tan x \cot x) dx = 0$$

Sol.11. (b)

$$\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$$

put $\tan^{-1} x = t$

$$\int_0^{\pi/4} t dt = \left[\frac{t^2}{2} \right]_0^{\pi/4} = \frac{\pi^2}{32}$$

Sol.12. (c)

x^3 and $\sin x$ both are odd functions so integration of both functions is zero.

Sol.13. (d)

$\sin^2 x$ is odd functions so integration of function is zero.

Sol.14. (b)

$$\int_0^2 e^{\log x} dx = \int_0^2 x dx = \left[\frac{x^2}{2} \right]_0^2 = 2$$

Sol.15. (c)

$$\int_1^2 \log x dx = [x \log x - x]_1^2 = 2 \log 2 - 1$$

$$= \log 4 - \log e = \log \left(\frac{4}{e} \right)$$

Sol.16. (b)

$$\int_{-\pi/2}^{\pi/2} x \sin x dx = 2 \int_0^{\pi/2} x \sin x dx$$

(even function)

$$= 2[-x \cos x + \sin x]_0^{\pi/2} = 2$$

Sol.17. (c)

$$I = \int_0^{\pi/2} \ln(\tan x) dx \quad \left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$I = \int_0^{\pi/2} \ln(\cot x) dx$$

$$2I = \int_0^{\pi/2} \ln(\tan x) dx + \int_0^{\pi/2} \ln(\cot x) dx$$

$$2I = \int_0^{\pi/2} \ln(\tan x \cot x) dx = 0$$

Sol.18. (a)

$$\int_0^1 \frac{e^{\tan^{-1} x} dx}{1+x^2}$$

put $\tan^{-1} x = t$

$$\int_0^{\pi/4} e^t dt = e^{\pi/4} - 1$$

Sol.19. (a)

$$I_1 = \int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}} = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x} dx}{\sqrt{\sin x} + \sqrt{\cos x}} \dots (i)$$

$$\left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$= \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x} dx}{\sqrt{\sin x} + \sqrt{\cos x}} = I_2 \dots (ii)$$

$$I_1 - I_2 = 0$$

Sol.20. (c)

(i) + (ii)

$$I + I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x} + \sqrt{\cos x} dx}{\sqrt{\sin x} + \sqrt{\cos x}} = \int_{\pi/6}^{\pi/3} 1 dx$$

$$2I = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

$$I = \frac{\pi}{12}$$

Sol.21. (d)

$$I = \int_0^{\pi} \ln(\sin x) dx$$

$$\left[\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx \text{ when } f(2a-x) = f(x) \right]$$

$$I = 2 \int_0^{\pi/2} \ln(\sin x) dx$$

$$\int_0^{\pi/2} \ln(\sin x) dx = \frac{I}{2}$$

Sol.22. (a)

$$\int_0^{\pi/2} \ln(\sin x) dx = \frac{I}{2}$$

$$\left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$\int_0^{\pi/2} \ln(\cos x) dx = \frac{I}{2}$$

Sol.23. (c)

$$\int_0^{\pi/2} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$$

$$= \int_0^{\pi/2} \frac{\sec^2 x dx}{a^2 + b^2 \tan^2 x} = \frac{1}{b^2} \int_0^{\pi/2} \frac{\sec^2 x dx}{\left(\frac{a}{b}\right)^2 + \tan^2 x}$$

put $\tan x = t$

$$= \frac{1}{b^2} \int_0^{\infty} \frac{dt}{\left(\frac{a}{b}\right)^2 + t^2} = \frac{1}{ab} \left[\tan^{-1} \left(\frac{x}{a/b} \right) \right]_0^{\infty} = \frac{\pi}{2ab}$$

Sol.24. (c)

$$I_1 = \int_0^{\pi} \frac{x dx}{1 + \sin x} \dots\dots\dots(i)$$

$$\left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$I_1 = \int_0^{\pi} \frac{(\pi-x) dx}{1 + \sin(\pi-x)} = \int_0^{\pi} \frac{(\pi-x) dx}{1 + \sin x} \dots\dots\dots(ii)$$

(i) + (ii)

$$2I_1 = \int_0^{\pi} \frac{\pi dx}{1 + \sin x} = \pi \int_0^{\pi} \frac{dx}{1 + \sin x} \dots\dots\dots(iii)$$

$$\left[\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx \text{ when } f(2a-x) = f(x) \right]$$

$$2I_1 = 2\pi \int_0^{\pi/2} \frac{dx}{1 + \sin x}$$

$$2I_1 = 2\pi \int_0^{\pi/2} \frac{dx}{1 + \cos\left(\frac{\pi}{2} - x\right)} = 2\pi \int_0^{\pi/2} \frac{dx}{2 \cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)}$$

$$I_1 = \pi \int_0^{\pi/2} \frac{\sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) dx}{2} = \frac{\pi}{2} \int_0^{\pi/2} \sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) dx$$

$$I_1 = \frac{\pi}{2} \left[\tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \right]_0^{\pi/2} \times (-2)$$

$$I_1 = \frac{\pi}{2} [0 - 1] \times (-2) = \pi$$

Sol.25. (a)

by equation(ii)

$$I_1 = \int_0^{\pi} \frac{(\pi-x) dx}{1 + \sin x} = \int_0^{\pi} \frac{x dx}{1 + \sin x} = \pi$$

Sol.26. (b)

by equation (iii)

$$2I_1 = \pi \int_0^{\pi} \frac{dx}{1 + \sin x}$$

$$\int_0^{\pi} \frac{dx}{1 + \sin x} = 2$$

Sol.27. (a)

$$I_1 = \int_0^{\pi} \frac{\sin 2x}{\sin x} dx = \int_0^{\pi} \frac{2 \sin x \cos x}{\sin x} dx$$

$$I_1 = \int_0^{\pi} 2 \cos x dx = 2[\sin x]_0^{\pi} = 0$$

Sol.28. (d)

$$I_m = \int_0^{\pi} \frac{\sin 2mx}{\sin x} dx = \int_0^{\pi} \frac{\sin 2m(\pi-x)}{\sin x} dx \dots\dots(i)$$

$$I_m = - \int_0^{\pi} \frac{\sin 2mx}{\sin x} dx \dots\dots(ii)$$

(i) + (ii)

$$2I_m = 0 \Rightarrow I_m = 0$$

$$\text{so } I_1 = I_2 = I_3 = 0$$

Sol.29. (a)

$$I_m = \int_0^{\pi} \frac{\sin 2mx}{\sin x} dx = \int_0^{\pi} \frac{\sin 2m(\pi-x)}{\sin x} dx \dots\dots(i)$$

$$I_m = - \int_0^{\pi} \frac{\sin 2mx}{\sin x} dx \dots\dots(ii)$$

(i) + (ii)

$$2I_m = 0 \Rightarrow I_m = 0$$

Sol.30. (a)

$$I_m = 0 \text{ so } I_{2m} > I_m \text{ is wrong}$$

Sol.31. (d)

$$\int_a^b \frac{x^7 + \sin x}{\cos x} dx = \int_{-b}^b \frac{x^7 + \sin x}{\cos x} dx = 0$$

$$\left[\int_{-a}^a f(x) dx = 0 \text{ if } f(-x) = -f(x) \right]$$

Sol.32. (a)

$$\int_{-1}^1 x |x| dx = 0$$

$$\left[\int_{-a}^a f(x) dx = 0 \text{ if } f(-x) = -f(x) \right]$$

Sol.33. (a)

$$\int_a^b \frac{|x|}{x} dx = \int_a^b 1 dx = b - a$$

Sol.34. (c)

$$A = \int_0^{\pi} \frac{\sin x dx}{\sin x + \cos x}$$

$$\left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$A = \int_0^{\pi} \frac{\sin(\pi-x) dx}{\sin(\pi-x) + \cos(\pi-x)} = \int_0^{\pi} \frac{\sin x dx}{\sin x - \cos x} = B$$

Sol.35. (b)

$$\int_0^{\pi} \frac{\sin x dx}{\sin x - \cos x} = \frac{1}{2} \int_0^{\pi} \frac{2 \sin x dx}{\sin x - \cos x}$$

$$= \frac{1}{2} \int_0^{\pi} \frac{\sin x + \cos x + \sin x - \cos x dx}{\sin x - \cos x}$$

$$= \frac{1}{2} \int_0^{\pi} \frac{(\sin x + \cos x) dx}{\sin x - \cos x} + \frac{1}{2} \int_0^{\pi} \frac{(\sin x - \cos x) dx}{\sin x - \cos x}$$

$$= \frac{1}{2} \int_0^{\pi} \frac{(\sin x + \cos x) dx}{\sin x - \cos x} + \frac{1}{2} \int_0^{\pi} 1 dx$$

$$= 0 + \pi/2$$

Sol.36. (c)

$$\int_0^{2\pi} \sin^5\left(\frac{x}{4}\right) dx = \int_0^{2\pi} \sin^4\left(\frac{x}{4}\right) \sin\left(\frac{x}{4}\right) dx$$

$$= \int_0^{2\pi} \left(1 - \cos^2 \frac{x}{4}\right) \left(1 - \cos^2 \frac{x}{4}\right) \sin\left(\frac{x}{4}\right) dx$$

$$\text{put } \cos\left(\frac{x}{4}\right) = t$$

$$= -4 \int_1^0 (1-t^2)(1-t^2) dt = -4 \int_1^0 (t^4 - 2t^2 + 1) dt$$

$$= 32/15$$

Sol.37. (d)

$$\int_0^{4\pi} |\cos x| dx = 8 \text{ (by graph)}$$

Sol.38. (b)

$$\int_{\frac{1}{3}}^{\frac{1}{2}} \left[\frac{1}{x} \right] dx = \int_{\frac{1}{3}}^{\frac{1}{2}} 2 dx = 2 \left[\frac{1}{2} - \frac{1}{3} \right] = \frac{1}{3}$$

Sol.39. (a)

$$\int_{\frac{1}{3}}^1 f(x) dx = \int_{\frac{1}{3}}^{1/2} xg(x) dx + \int_{1/2}^1 xg(x) dx$$

$$= \int_{\frac{1}{3}}^{1/2} x(2) dx + \int_{1/2}^1 x \cdot 1 dx$$

$$= \left[x^2 \right]_{1/3}^{1/2} + \left[\frac{x^2}{2} \right]_{1/2}^1 = \frac{5}{36} + \frac{3}{8} = \frac{37}{72}$$

Sol.40. (c)

$$\int_{-2}^2 x dx - \int_{-2}^2 [x] dx$$

$$= 0 - \int_{-2}^{-1} [x] dx - \int_{-1}^0 [x] dx - \int_0^1 [x] dx - \int_1^2 [x] dx$$

$$= - \int_{-2}^{-1} (-2) dx - \int_{-1}^0 (-1) dx - \int_0^1 (0) dx - \int_1^2 (1) dx$$

$$= 2 + 1 + 1 = 4$$

Sol.41. (b)

$$\int_{-2}^0 f(x) dx + \int_0^5 f(x) dx = 4 \dots\dots(i)$$

$$\text{given } \int_0^5 1 dx + \int_0^5 \{f(x)\} dx = 7,$$

$$= 5 + \int_0^5 \{f(x)\} dx = 7$$

$$= \int_0^5 \{f(x)\} dx = 2$$

now by (i)

$$\int_{-2}^0 f(x) dx + 2 = 4 \Rightarrow \int_{-2}^0 f(x) dx = 2$$

$$\int_0^5 \{1 + f(x)\} dx \text{ then what is } \int_{-2}^0 f(x) dx \text{ equal to?}$$

Sol.42. (c)

$$a_{2n} = \int_0^{\pi} \frac{\sin^2\{(2n+1)x\}}{\sin 2x} dx \dots\dots\dots(i)$$

$$a_{2n} = \int_0^{\pi} \frac{\sin^2\{(2n+1)(\pi-x)\}}{\sin 2(\pi-x)} dx = \int_0^{\pi} \frac{\sin^2\{(2n+1)x\}}{-\sin 2x} dx$$

.....(ii)

add both equations

$$a_{2n} = 0$$

all terms are zero so sequence is in AP with

difference 0

same as above

$$a_{2n+1} = 0$$

all terms are zero so sequence is in AP with

difference 0

Sol.43. (b)

$$a_{n-1} = a_{n-4}$$

Sol.44. (c)

$$f'\left(\frac{1}{x}\right) = -x^3 f'(x)$$

divide by x^2

$$-\frac{1}{x^2} f'\left(\frac{1}{x}\right) = x f'(x)$$

integrate both sides

$$-\int_{-1}^1 \frac{1}{x^2} f'\left(\frac{1}{x}\right) dx = \int_{-1}^1 x f'(x) dx$$

let $1/x = t$ in left hand side and integrate right hand side by parts

$$-\int_{-1}^1 f'(t) dt = [xf(x)]_{-1}^1 - \int_{-1}^1 f(x) dx$$

$$[f(t)]_{-1}^1 = [xf(x)]_{-1}^1 - \int_{-1}^1 f(x) dx$$

$$f(1) - f(-1) = f(1) + f(-1) - \int_{-1}^1 f(x) dx$$

$$\int_{-1}^1 f(x) dx = 2f(-1)$$

Sol.45. (b)

$$\int_0^{\pi/2} \frac{dx}{3\cos x + 5} = \int_0^{\pi/2} \frac{dx}{3 \left[\frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right] + 5}$$

$$= \int_0^{\pi/2} \frac{[1 + \tan^2 \frac{x}{2}] dx}{3[1 - \tan^2 \frac{x}{2}] + 5[1 + \tan^2 \frac{x}{2}]}$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{\sec^2 \frac{x}{2} dx}{\tan^2 \frac{x}{2} + 2^2}$$

put $\tan \frac{x}{2} = t$

$$= \int_0^1 \frac{dx}{t^2 + 2^2} = \frac{1}{2} \left[\tan^{-1} \left(\frac{t}{2} \right) \right]_0^1 = \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right)$$

$$= \frac{1}{2} \cot^{-1} 2$$

$$k = \frac{1}{2}$$

Sol.46. (d)

$$\int_1^3 [1 - x^4] dx = \int_1^3 (1 - x^4) dx$$

$$\left[\frac{x^5}{5} - x \right]_1^3 = \left[\frac{3^5}{5} - 3 \right] - \left[\frac{1}{5} - 1 \right] = \frac{232}{5}$$

Sol.47. (b)

$$\int_0^{\pi/2} \frac{d\theta}{1 + \cos \theta} = \int_0^{\pi/2} \frac{d\theta}{2\cos^2 \left(\frac{\theta}{2} \right)} = \frac{1}{2} \int_0^{\pi/2} \sec^2 \left(\frac{\theta}{2} \right) d\theta$$

$$\frac{1}{2} \int_0^{\pi/2} 2 \tan \left(\frac{\theta}{2} \right) d\theta = 1$$

Sol.48. (b)

$$I = \int_0^a f(x) g(x) dx = \int_0^a f(a-x) g(a-x) dx$$

$$I = \int_0^a f(x) [2 - g(x)] dx$$

$$I = \int_0^a [2f(x) - f(x)g(x)] dx$$

$$I = \int_0^a 2f(x) dx - \int_0^a f(x)g(x) dx$$

$$I = \int_0^a 2f(x) dx - I$$

$$2I = \int_0^a 2f(x) dx \Rightarrow I = \int_0^a f(x) dx$$

Sol.49. (b)

$$\int_{e^{-1}}^{e^2} \left| \frac{\ln x}{x} \right| dx$$

$$dx = \int_{e^{-1}}^1 - \left(\frac{\ln x}{x} \right) dx + \int_1^{e^2} \left(\frac{\ln x}{x} \right) dx$$

$$= -\frac{1}{2} [(\log x)^2]_{1/e}^1 + \frac{1}{2} [(\log x)^2]_1^{e^2}$$

$$= \frac{1}{2} + \frac{4}{2} = \frac{5}{2}$$

Sol.50. (a)

$$\int_0^{2x} \sqrt{1 + \sin \frac{x}{2}} dx = \int_0^{2x} \left| \sin \frac{x}{4} + \cos \frac{x}{4} \right| dx$$

$$= 4 \left[-\cos \frac{x}{2} + \sin \frac{x}{4} \right]_0^{2x} = 8$$

Sol.51. (d)

$$\int_0^{x/4} (\sqrt{\tan x} + \sqrt{\cot x}) dx = \int_0^{x/4} \left(\frac{\sin x + \cos x}{\sqrt{\sin x \cdot \cos x}} \right) dx$$

$$= \sqrt{2} \int_0^{x/4} \left(\frac{\sin x + \cos x}{\sqrt{1 + \sin^2 x + \cos^2 x - 2 \sin x \cdot \cos x}} \right) dx$$

$$= \sqrt{2} \int_0^{x/4} \left(\frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} \right) dx$$

put $\sin x - \cos x = t$

then $(\sin x + \cos x) dx = dt$

$$= \sqrt{2} \int_{-1}^0 \left(\frac{dt}{\sqrt{1 - t^2}} \right) = \sqrt{2} [\sin^{-1} x]_{-1}^0 = \frac{\pi}{\sqrt{2}}$$

Sol.52. (a)

$$\int_0^1 x(1-x)^9 dx = \int_0^1 x^9(1-x) dx = \int_0^1 (x^9 - x^{10}) dx$$

$$= \left[\frac{x^{10}}{10} - \frac{x^{11}}{11} \right]_0^1 = \left[\frac{1}{10} - \frac{1}{11} \right] = \frac{1}{110}$$

Sol.53. (a)

$$\int_a^b x^3 dx = \left[\frac{x^4}{4} \right]_a^b = \left[\frac{a^4 - b^4}{4} \right] = 0$$

$$\Rightarrow a^4 - b^4 = 0 \Rightarrow a = \pm b$$

$$\int_a^b x^2 dx = \left[\frac{x^3}{3} \right]_a^b = \left[\frac{a^3 - b^3}{3} \right] = \frac{2}{3}$$

$$\Rightarrow a^3 = b^3 + 2$$

by these equations $a = 1$ and $b = -1$

Sol.54. (c)

odd function so by direct property answer is zero.

Sol.55. (a)

$$\int_0^{\sqrt{2}} [x^2] dx = \int_0^1 0 dx + \int_1^{\sqrt{2}} 1 dx = \sqrt{2} - 1$$

Sol.56. (b)

$$\int_1^c x \ln x dx = \left[\ln x \cdot \frac{x^2}{4} \right]_1^c - \int_1^c \frac{1}{x} \cdot \frac{x^2}{2} dx$$

$$= \frac{e^2}{2} - \left[\frac{x^2}{4} \right]_1^e = \frac{e^2 + 1}{4}$$

Sol.57. (b)

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$\int_0^{\pi} e^x \sin x dx = \left[\frac{e^x}{2} (\sin x - \cos x) \right]_0^{\pi} = \frac{e^{\pi} + 1}{2}$$

Sol.58. (d)

$$\int_{-1}^1 \left\{ \frac{d}{dx} \left(\tan^{-1} \frac{1}{x} \right) \right\} dx = \left[\tan^{-1} \frac{1}{x} \right]_{-1}^1$$

$$= \tan^{-1} 1 + \tan^{-1} 1 = \pi/2$$

Sol.59. (d)

$$\int_2^8 |x - 5| dx = -\int_2^5 (x - 5) dx + \int_5^8 (x - 5) dx$$

$$= - \left[\frac{x^2}{2} - 5x \right]_2^5 + \left[\frac{x^2}{2} - 5x \right]_5^8$$

$$= - \left[\frac{25}{2} - 25 - 2 + 10 \right] + \left[32 - 40 - \frac{25}{2} + 25 \right] = 9$$

Sol.60. (b)

$$\int_a^b [x] dx + \int_a^b [-x] dx = \int_a^b k dx + \int_a^b (-1 - k) dx$$

$$= \int_a^b (k - 1 - k) dx = \int_a^b (-1) dx = a - b$$

Sol.61. (b)

$$\int_0^{\pi/2} |\sin x - \cos x| dx$$

$$= - \int_0^{\pi/4} (\sin x - \cos x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

$$= - [-\cos x - \sin x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2}$$

$$= - \left[-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 1 \right] + \left[-1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right]$$

$$= 2(\sqrt{2} - 1)$$

Sol.62. (b)

$$\int_0^{\pi/2} e^{\sin x} \cos x dx$$

$$= \left[e^{\sin x} \right]_0^{\pi/2} = e^{\sin \pi/2} e^0 = e - 1$$

Sol.63. (b)

$$I_1 = \int_0^{\pi} \frac{x dx}{1 + \sin x} \dots \dots \dots (i)$$

$$\left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$I_1 = \int_0^{\pi} \frac{(\pi - x) dx}{1 + \sin(\pi - x)} = \int_0^{\pi} \frac{(\pi - x) dx}{1 + \sin x} \dots \dots \dots (ii)$$

$$(i) + (ii)$$

$$2I_1 = \int_0^{\pi} \frac{\pi dx}{1 + \sin x} = \pi \int_0^{\pi} \frac{dx}{1 + \sin x}$$

$$\left[\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx \text{ when } f(2a-x) = f(x) \right]$$

$$2I_1 = 2\pi \int_0^{\pi/2} \frac{dx}{1 + \sin x}$$

$$2I_1 = 2\pi \int_0^{\pi/2} \frac{dx}{1 + \cos\left(\frac{\pi-x}{2}\right)} = 2\pi \int_0^{\pi/2} \frac{dx}{2 \cos^2\left(\frac{\pi-x}{4}\right)}$$

$$I_1 = \pi \int_0^{\pi/2} \frac{\sec^2\left(\frac{\pi-x}{4}\right) dx}{2} = \frac{\pi}{2} \int_0^{\pi/2} \sec^2\left(\frac{\pi-x}{4}\right) dx$$

$$I_1 = \frac{\pi}{2} \left[\tan\left(\frac{\pi-x}{4}\right) \right]_0^{\pi/2} \times (-2)$$

$$I_1 = \frac{\pi}{2} [0 - 1] \times (-2) = \pi$$

Sol.64. (d)

$$I_2 = \int_0^{\pi} \frac{(\pi-x) dx}{1 - \sin(\pi+x)}$$

$$\left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$I_2 = \int_0^{\pi} \frac{x dx}{1 + \sin x} = I_1$$

by previous question

$$I_1 = \pi$$

$$I_1 + I_2 = 2\pi$$

Sol.65. (b)

$$\int_0^{\pi/4} (\tan^3 x + \tan x) dx$$

$$= \int_0^{\pi/4} \tan x (\tan^2 x + 1) dx$$

$$= \int_0^{\pi/4} \tan x (\sec^2 x) dx$$

put $\tan x = t$

$$= \int_0^1 t dt = \left[\frac{t^2}{2} \right]_0^1 = \frac{1}{2}$$

Sol.66. (c)

$$I = \int_0^a \frac{f(a-x)}{f(x) + f(a-x)} dx$$

$$\left[\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$I = \int_0^a \frac{f(x)}{f(x) + f(a-x)} dx$$

$$2I = 0$$

$$I = 0$$

Sol.67. (a)

$$\int_0^a [f(x) + f(-x)] dx = \int_{-a}^a g(x) dx$$

$$\int_0^a f(x) dx + \int_0^a f(-x) dx = \int_{-a}^0 g(x) dx + \int_0^a g(x) dx$$

$$\text{Solving } \int_{-a}^0 g(x) dx \text{ individually}$$

$$\int_{-a}^0 g(x) dx$$

Put $x = -t$, $dx = -dt$

$$\int_{-a}^0 g(x) dx = - \int_a^0 g(-t) dx = \int_0^a g(-x) dx$$

$$\int_0^a f(x) dx + \int_0^a f(-x) dx = \int_0^a g(-x) dx + \int_0^a g(x) dx$$

then $g(x) = f(x)$

Sol.68. (d)

$$\int_2^{10} \frac{f'(x)}{f(x)} dx$$

$$= [\log f(x)]_2^{10}$$

$$= \log f(10) - \log f(2)$$

$$= \log \frac{f(10)}{f(2)}$$

$$= \log \frac{2^{10}}{2^2} = \log 2^8$$

$$= 8 \log 2$$

Sol.69. (d)

$$\int_{-2}^0 f(x) dx = k \quad (k \in \mathbb{R})$$

Case 1: If graph of $f(x)$ lie above the x axis then $|f(x)| = f(x)$

$$\int_{-2}^0 |f(x)| dx = k \quad (k \in \mathbb{R})$$

Case 2: If graph of $f(x)$ lie below the x axis

$|f(x)| = -f(x)$

$$\int_{-2}^0 |f(x)| dx = k \quad (k \in \mathbb{R}^+)$$

Case 3: If graph of $f(x)$

Cuts the x axis

Then Area of $|f(x)|$

Will be greater than $f(x)$

$$\int_{-2}^0 |f(x)| dx > k$$

So answer is equal to or greater than k .

Sol.70. (b)

$$\int_1^4 f'(x) dx = [f(x)]_1^4 = f(4) - f(1) = 0$$

Sol.71. (c)

$$\int_0^{\pi/2} e^{\log \cos x} dx = \int_0^{\pi/2} \cos x dx = [\sin x]_0^{\pi/2} = 1$$

Sol.72. (a)

$$I = \int_0^{\pi} \log \left(\tan \frac{x}{2} \right) dx \quad \dots (1)$$

$$f(x) = f(a+b-x)$$

$$I = \int_0^{\pi} \log \left(\cot \frac{x}{2} \right) dx \quad \dots (2)$$

By adding both equations

$$2I = \int_0^{\pi} \left[\log \left(\tan \frac{x}{2} \right) + \log \left(\cot \frac{x}{2} \right) \right] dx$$

$$= \int_0^{\pi} \log \left(\tan \frac{x}{2} \cdot \cot \frac{x}{2} \right) dx$$

$$= \int_0^{\pi} \log 1 dx = 0$$

Sol.73. (b)

$$\int_0^{\pi/4} \frac{dx}{(\sin x + \cos x)^2}$$

$$= \int_0^{\pi/4} \frac{\sec^2 x}{(\tan x + 1)^2} dx$$

Put $\tan x = t$

$\sec^2 x dx = dt$ and change limit

$$\Rightarrow \int_0^1 \frac{1}{(t+1)^2} dt$$

$$= \left[-\frac{1}{t+1} \right]_0^1$$

$$= -\frac{1}{2} + 1 = \frac{1}{2}$$

Sol.74. (b)

$$\int_{-2}^{-1} \frac{x}{|x|} dx$$

From -2 to -1

$$\Rightarrow \frac{x}{|x|} = -1$$

$$\Rightarrow \int_{-2}^{-1} (-1) dx = (-x)_{-2}^{-1}$$

$$= -2 - (-1)$$

Sol.75. (b)

$$I = \int_0^1 \ln \left(\frac{1}{x} - 1 \right) dx = \int_0^1 \ln \left(\frac{1-x}{x} \right) dx \quad \dots (i)$$

$$\int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$I = \int_0^1 \ln \left(\frac{x}{1-x} \right) dx \quad \dots (ii)$$

Add both equations

$$2I = \int_0^1 \ln \left(\frac{1-x}{x} \right) + \ln \left(\frac{x}{1-x} \right) dx$$

$$2I = \int_0^1 \ln \left(\frac{1-x}{x} \right) \left(\frac{x}{1-x} \right) dx$$

$$2I = \int_0^1 \log 1 dx = 0$$

Sol.76. (d)

$$I = \int_0^{20\pi} (\sin^4 x + \cos^4 x) dx = 20 \int_0^{\pi} (\sin^4 x + \cos^4 x) dx$$

$$= 40 \int_0^{\pi/2} (\sin^4 x + \cos^4 x) dx = 40k$$

Sol.77. (a)

$$\int_{-\pi/2}^{\pi/2} (e^{\cos x} \sin x + e^{\sin x} \cos x) dx$$

Here $e^{\cos x} \sin x$ is an odd function so

$$\int_{-\pi/2}^{\pi/2} (e^{\cos x} \sin x) dx = 0$$

And

$$\int_{-\pi/2}^{\pi/2} (e^{\sin x} \cos x) dx = (e^{\sin x})_{-\pi/2}^{\pi/2} = e - \frac{1}{e} = \frac{e^2 - 1}{e}$$

Solution for three questions

$$f(x) = Pe^x + Qe^{2x} + Re^{3x}$$

$$f(0) = P + Q + R = 6 \quad \dots(1)$$

$$f'(x) = Pe^x + 2Qe^{2x} + 3Re^{3x}$$

$$f'(\log 3) = 3P + 18Q + 81R = 282$$

$$\Rightarrow P + 6Q + 27R = 94 \quad \dots(2)$$

$$\int_0^{\log 2} f(x) dx = \left[Pe^x + \frac{Qe^{2x}}{2} + \frac{Re^{3x}}{3} \right]_0^{\log 2}$$

$$= 2P + 2Q + 8/3 R - P - Q/2 - R/3$$

$$= P + \frac{3}{2}Q + \frac{7}{3}R = 11 \quad \dots(3)$$

By solving above three equations

$$P=1, Q=2, R=3$$

Sol.78. (b)

By above solution $Q=2$

Sol.79. (c)

By above solution $R=3$

Sol.80. (d)

By above solution

$$f'(0) = P + 2Q + 3R$$

$$= 1 + 4 + 9 = 14$$

Sol.81. (d)

$$I_1 = \int_0^{\pi} \frac{x}{1 + \cos^2 x} dx$$

...(1)

$$\int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$I_1 = \int_0^{\pi} \frac{\pi - x}{1 + \cos^2 x} dx$$

...(2)

(1)+(2)

$$2I_1 = \pi \int_0^{\pi} \frac{x}{1 + \cos^2 x} dx$$

$$2I_1 = 2\pi \int_0^{\pi/2} \frac{1}{1 + \sin^2 x} dx$$

$$I_1, II, \int_0^{\pi/2} \frac{1}{1 + \sin^2 x} dx$$

...(3)

$$I_2 = \int_0^{\pi} \frac{1}{1 + \sin^2 x} dx = 2 \int_0^{\pi/2} \frac{1}{1 + \sin^2 x} dx$$

...(4)

By (3) & (4)

$$\int_0^{\pi/2} \frac{1}{1 + \sin^2 x} dx = \frac{I_1}{\pi} = \frac{I_2}{2}$$

...(5)

$$\Rightarrow \frac{I_1}{I_2} = \frac{\pi}{2} \Rightarrow \frac{I_1 + I_2}{I_1 - I_2} = \frac{\pi + 2}{\pi - 2}$$

By equation (3)

Sol.82. (d)

$$I_1 = \pi \int_0^{\pi/2} \frac{1}{1 + \sin^2 x} dx = \pi \int_0^{\pi/2} \frac{\sec^2 x dx}{1 + 2 \tan^2 x}$$

But $\tan x = t$
 $\sec^2 x dx = dt$

$$I_1 = \frac{\pi}{2} \int_0^{\infty} \frac{dt}{\left(\frac{1}{\sqrt{2}}\right)^2 + t^2}$$

$$= \frac{\pi}{2} \sqrt{2} \left[\tan^{-1}(\sqrt{2}t) \right]_0^{\infty} = \frac{\pi^2}{2\sqrt{2}}$$

$$I_1 = \frac{\pi^2}{2\sqrt{2}}$$

$$I_1^2 = \frac{\pi^4}{8}$$

$$8 I_1^2 = \pi^4$$

Sol.83. (a)

By equation (5)

$$I_2 = 2/\pi I_1$$

$$= \frac{2}{\pi} \cdot \frac{\pi^2}{2\sqrt{2}}$$

$$I_2 = \pi/\sqrt{2}$$

Sol.84. (a)

$$\int_a^b \frac{|x|}{x} dx$$

for $a < 0 < b$

$$\int_a^0 (-1) dx + \int_0^b 1 dx$$

$$= (-x)_a^0 + (x)_0^b$$

$$= a + b$$

Sol.85. (b)

$$\int_a^b \frac{|x|}{x} dx$$

for $a < 0 < b$

$$\int_a^0 (-1) dx = -(x)_a^0$$

$$= -[b-a]$$

$$= a - b$$

Sol.86. (b)

$$I = \int_{-2\pi}^{2\pi} \frac{\sin^4 x + \cos^4 x}{1 + 3^x} dx \dots\dots\dots(i)$$

$$\int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$I = \int_{-2\pi}^{2\pi} \frac{\sin^4 x + \cos^4 x}{1 + 3^{-x}} dx \dots\dots\dots(ii)$$

add (i) and (ii)

$$2I = \int_{-2\pi}^{2\pi} \frac{(1 + 3^x)(\sin^4 x + \cos^4 x)}{1 + 3^x} dx$$

$$2I = \int_{-2\pi}^{2\pi} \sin^4 x + \cos^4 x dx$$

$$2I = 2 \int_0^{2\pi} \sin^4 x + \cos^4 x dx$$

$$I = \int_0^{2\pi} \sin^4 x + \cos^4 x dx$$

$$I = 4 \int_0^{\pi/2} (\sin^4 x + \cos^4 x) dx$$

$$I = 4 \int_0^{\pi/2} (1 - 2\sin^2 x \cos^2 x) dx$$

$$I = 4 \int_0^{\pi/2} \left(1 - \frac{\sin^2 2x}{2} \right) dx = 4 \int_0^{\pi/2} \left(1 - \frac{1 - \cos 4x}{4} \right) dx$$

$$I = 4 \int_0^{\pi/2} \left(\frac{3}{4} + \frac{\cos 4x}{5} \right) dx = 4 \left(\frac{3x}{4} + \frac{\sin 4x}{16} \right)_{\pi/2}^{\pi/2}$$

$$I = 4 \int_0^{\pi/2} \left(\frac{3}{4} + \frac{\cos 4x}{5} \right) dx = 4 \left(\frac{3\pi}{8} \right) = \frac{3\pi}{2}$$

$$I = \int_0^{2\pi} \sin^4 x + \cos^4 x dx = \frac{3\pi}{2}$$

$$I = 2 \int_0^{\pi} \sin^4 x + \cos^4 x dx = \frac{3\pi}{2}$$

$$\int_0^{\pi} \sin^4 x + \cos^4 x dx = \frac{3\pi}{4}$$

Sol.87. (c)

By above solution

$$I = \int_0^{2\pi} \sin^4 x + \cos^4 x dx = \frac{3\pi}{2}$$

Sol.88. (d)

$$\begin{aligned} 8\pi \int_0^{\pi} |\sin x| dx &= 8 \int_0^{\pi} |\sin x| dx \\ &= 8 \int_0^{\pi} \sin x dx = 8 \times 2 = 16 \end{aligned}$$

Sol.89. (a)

$$\begin{aligned} \int_{-1}^0 |x-1| dx &= \int_{-1}^0 \{-(x-1)(-1)\} dx \\ &= \int_{-1}^0 (x-1) dx = \left[\frac{x^2}{2} - x \right]_{-1}^0 = -\frac{3}{2} \end{aligned}$$

Sol.90. (d)

$$\begin{aligned} \int_0^2 |x-1| dx &= \int_0^1 |x-1| dx + \int_1^2 |x-1| dx \\ &= \int_0^1 \{-(x-1) \times 0\} dx + \int_1^2 \{(x-1) \times 1\} dx \\ &= 0 + \left[\frac{x^2}{2} - x \right]_1^2 = \frac{1}{2} \end{aligned}$$

Sol.91. (a)

$$3f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x} + 1 \quad \dots\dots\dots(i)$$

replace x with $1/x$

$$3f\left(\frac{1}{x}\right) + f(x) = x + 1 \quad \dots\dots\dots(ii)$$

by solving above equations

$$f(x) = \frac{3}{8x} - \frac{x}{8} + \frac{1}{4}$$

$$= \left[\frac{3}{8} \log 2 - \frac{4}{16} + \frac{2}{4} - 0 + \frac{1}{16} - \frac{1}{4} \right]$$

$$= \left[3 \log 2 + \frac{1}{2} \right] = 3 \log 2 + \frac{1}{2} \log e$$

$$= \log 8 + \log \sqrt{e} = \log(8\sqrt{e})$$

Sol.92. (b)

$$I = \int_{-1}^1 (3 \sin x - \sin 3x) \cos^2 x dx$$

$$I = \int_{-1}^1 (4 \sin^3 x) \cos^2 x dx = 0$$

because above function is odd function

Sol.93. (d)

$y = f(x) = e^{-x}$ is positive for all real value of x
so $|f(x)| = f(x)$

than $p = q$

Sol.94. (a)

$$I = \int_0^{\frac{\pi}{2}} \frac{a + \sin x}{2a + \sin x + \cos x} dx \quad \dots\dots(i)$$

$$\text{apply } \int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$I = \int_0^{\frac{\pi}{2}} \frac{a + \cos x}{2a + \sin x + \cos x} dx \quad \dots\dots(ii)$$

add (i) and (ii)

$$2I = \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Sol.95. (d)

$$\phi(a) = \int_a^{a+100\pi} |\sin x| dx$$

$$\phi(a) = \int_0^{100\pi} |\sin x| dx$$

given function is periodic function

$$\phi(a) = 100 \int_0^{\pi} |\sin x| dx$$

$$\phi(a) = 100 \int_0^{\pi} \sin x dx = 200$$

Sol.96.

by above solution

$$\phi(a) = 200$$

$$\phi'(a) = 0$$

Sol.97. (d)

$$f(x) = |x^2 - x - 2|$$

$$f(x) = |(x-2)(x+1)|$$

$f(x)$ is positive for $x < -1$ and $x > 2$

$f(x)$ is negative for $-1 < x < 2$

$$\int_0^2 f(x) dx = - \int_0^2 (x^2 - x - 2) dx$$

$$= - \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x \right]_0^2$$

$$= - \left[\frac{8}{3} - \frac{4}{2} - 4 \right] = \frac{10}{3}$$

Sol.98. (b)

$$\int_1^3 f(x) dx = \int_1^2 f(x) dx + \int_2^3 f(x) dx$$

$$= - \int_1^2 (x^2 - x - 2) dx + \int_2^3 (x^2 - x - 2) dx$$

$$= - \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x \right]_1^2 + \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x \right]_2^3$$

$$= - \left[\frac{8}{3} - \frac{4}{2} - 4 - \frac{1}{3} + \frac{1}{2} + 2 \right] + \left[\frac{27}{3} - \frac{9}{2} - 6 - \frac{8}{3} + \frac{4}{2} + 4 \right]$$

$$= 3$$

Sol.99. (b)

$$f(t) = \ln(t + \sqrt{1+t^2})$$

$$f(-t) = \ln(-t + \sqrt{1+t^2})$$

$$f(t) + f(-t) = \ln(t + \sqrt{1+t^2}) + \ln(-t + \sqrt{1+t^2})$$

$$f(t) + f(-t) = 0$$

$$f(-t) = -f(t) \quad \dots\dots(i)$$

$$g(t) = \tan(f(t))$$

$$g(-t) = \tan(f(-t)) = \tan(-f(t))$$

$$= -\tan(f(t)) = -g(t)$$

so $g(t)$ is and odd function

and for odd functions

$$\int_{-a}^a g(t) dt = 0$$

Sol.100. (c)

$$\int_0^{\pi/2} \frac{1}{g(x)} dx$$

$$= \int_0^{\pi/2} \frac{dx}{\sin x + \cos x + 1}$$

$$= \int_0^{\pi/2} \frac{dx}{\frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} + \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} + 1}$$

$$= \int_0^{\pi/2} \frac{\left(1 + \tan^2 \frac{x}{2}\right) dx}{2 \tan \frac{x}{2} + 1 - \tan^2 \frac{x}{2} + 1 + \tan^2 \frac{x}{2}}$$

$$= \int_0^{\pi/2} \frac{\left(\sec^2 \frac{x}{2}\right) dx}{2 + 2 \tan \frac{x}{2}}$$

$$\text{let } 2 + 2 \tan \frac{x}{2} = t$$

$$= \ln \left[2 + 2 \tan \frac{x}{2} \right]_0^{\pi/2} = \ln(2 + 2 - 2) = \ln 2$$

Sol.101. (c)

$$I = \int_0^{\pi/2} \frac{f(x)}{g(x)} dx$$

$$I = \int_0^{\pi/2} \frac{\sin x dx}{\sin x + \cos x + 1} \quad \dots\dots(i)$$

$$\text{apply } \int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$I = \int_0^{\pi/2} \frac{\cos x dx}{\sin x + \cos x + 1} \quad \dots\dots(ii)$$

add both equations

$$2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x + 1} dx$$

add and subtract 1

$$2I = \int_0^{\pi/2} \frac{(\sin x + \cos x + 1 - 1) dx}{\sin x + \cos x + 1}$$

$$2I = \int_0^{\pi/2} 1 - \frac{1}{\sin x + \cos x + 1} dx$$

$$2I = \int_0^{\pi/2} 1 dx - \int_0^{\pi/2} \frac{1}{\sin x + \cos x + 1} dx$$

$$2I = \frac{\pi}{2} - \ln 2$$

$$I = \frac{\pi}{4} - \frac{\ln 2}{2}$$

Sol.102. (b)

$$\int_{\sqrt{2}}^{\sqrt{3}} f(x) dx = \int_{\sqrt{2}}^{\sqrt{3}} [x^2] dx = \int_{\sqrt{2}}^{\sqrt{3}} 2x dx$$

$$[2x]_{\sqrt{2}}^{\sqrt{3}} = 2\sqrt{3} - 2\sqrt{2}$$

Sol.103. (a)

$$\int_{\sqrt{2}}^2 f(x) dx = \int_{\sqrt{2}}^{\sqrt{3}} f(x) dx + \int_{\sqrt{3}}^2 f(x) dx$$

$$\Rightarrow \int_{\sqrt{2}}^{\sqrt{3}} [x^2] dx + \int_{\sqrt{3}}^2 [x^2] dx \Rightarrow \int_{\sqrt{2}}^{\sqrt{3}} 2x dx + \int_{\sqrt{3}}^2 3x dx$$

$$\Rightarrow [2x]_{\sqrt{2}}^{\sqrt{3}} + [3x]_{\sqrt{3}}^2 = 2\sqrt{3} - 2\sqrt{2} + 6 - 3\sqrt{3} = 6 - \sqrt{3} - 2\sqrt{2}$$

Sol.104. (c)

$$f(x) = x - [x]$$

This function is called fractional part of x whose time period is 1.

$$\int_n^{n+1} (x - [x]) dx = \int_0^1 x dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}$$

Sol.105. (a)

Let

$$\ln x = t$$

$$x = e^t$$

$$dx = e^t dt$$

$$I_1 = \int_e^{e^2} \frac{dx}{\ln x} = \int_1^2 \frac{e^t}{t} dt = \int_1^2 \frac{e^x}{x} dx = I_2$$

$$\text{So } I_1 - I_2 = 0$$